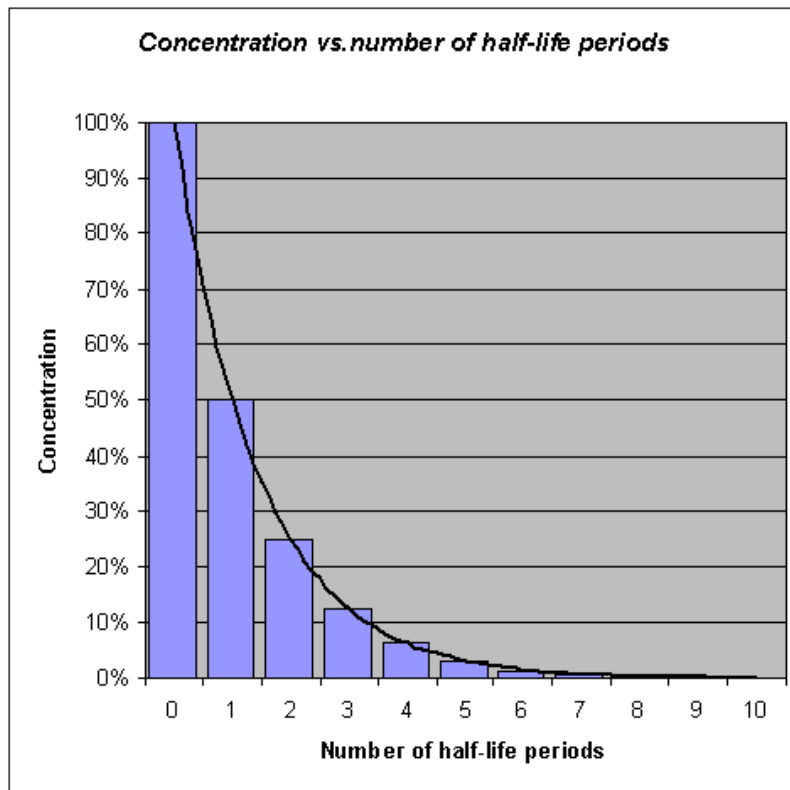




*Figure 1.6 Half-Life Affecting Blood Concentration of Medication Over Time*

Half-life affects the duration of the therapeutic effect of a medication. Many factors can influence half-life. For example, liver disease can prolong half-life if it is no longer effectively metabolizing the medication. Information about half-life of a specific medication can be found in evidence-based medication references. For example, in the “Clinical Pharmacology” section of the [DailyMed](#) reference for [furosemide](#), the half-life is approximately two hours.

Depending on whether a drug is metabolized and eliminated by the kidneys or liver, impairment in either of these systems can significantly alter medication dosing, frequency of doses, anticipated therapeutic effect, and even whether a particular medication can be used at all. Nurses must work with other members of the health care team to prevent drug interactions that could significantly affect a client’s health and well-being. Nurses must be alert for signs of a toxic buildup of metabolites or active drugs, particularly if the client has liver or kidney disease, so that they can alert the health care